

Sara Kia

AI | Computer Vision | Software Engineering

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GitHub: github.com/sarakia2050 | Portfolio Website: <https://sarakia2050.github.io/> | Live Project: ApplyMate Agent – <https://v0-applymate-agent.vercel.app/>

PROFESSIONAL SUMMARY

AI and Data Science professional with a Master's degree in Artificial Intelligence and a Bachelor's degree in Physics. Experienced in Python, machine learning, data analysis, computer vision, image processing, and robotics-related projects.

Currently strengthening software engineering skills at 42 Heilbronn, with a focus on C programming under Linux, Git, algorithms, problem solving, and project-based development.

Interested in practical roles in Data Science, Data Engineering, Machine Learning, Computer Vision, AI applications, and Software Engineering. Open to internships, working student positions, junior roles, part-time, full-time, and project-based opportunities in Germany.

TECHNICAL SKILLS

Programming:	Python, C, SQL, basic Java
Data & ML:	Pandas, NumPy, Scikit-learn, Statistics, Data Cleaning, Data Visualization, Machine Learning
Computer Vision:	OpenCV, Image Processing, Object Detection, OCR basics, Image Preprocessing
Software Engineering:	Linux, Git/GitHub, Algorithms, Memory Management, Pointers, Makefile basics
AI & Robotics:	SLAM, Evolutionary Algorithms, AI Agents, Robotics Vision
Tools:	VS Code, Jupyter Notebook, Google Colab, GitHub, MATLAB
Current Focus:	Computer Vision, Machine learning, image processing, AI Agents

SELECTED PROJECTS – AI, DATA & COMPUTER VISION

ApplyMate Agent – AI Job Search Agent Prototype – Built a web-based AI agent prototype that analyzes uploaded CVs and suggests matching career paths and job opportunities.

OpenCV Image Processing Project – Implemented an image processing pipeline including grayscale conversion, thresholding, Gaussian blur, resizing, cropping, annotation, and edge detection.

OpenCV-based Object Detection – Developed a basic computer vision pipeline for detecting and analyzing objects in images using Python and OpenCV.

Persian Handwriting Recognition – Worked on a machine learning and computer vision project for recognizing handwritten Persian text using image preprocessing and classification techniques.

License Plate Recognition – Image Processing Project – Developed an image processing workflow to detect vehicles passing through a red light and extract license plate regions for municipal traffic monitoring.

SLAM Project with Evolutionary Algorithms – Worked on mapping and localization in robotics using sensor data and evolutionary algorithms.

SpaceX Launch Prediction Project – Built a machine learning model using structured real-world data, including cleaning, visualization, feature selection, and evaluation.

Libft – C Programming Project at 42 Heilbronn – Built a custom C library by re-implementing core standard functions, focusing on memory management, pointers, string handling, and Makefile usage.

PROFESSIONAL EXPERIENCE

Programmer – Amiri & Daraei Company | 2013 – 2021

- Worked on software development, technical problem solving, and practical programming tasks.
- Applied programming, analytical thinking, system design skills, and data-related workflows in practical projects.
- Gained experience in Python, data-related workflows, software logic, and technical documentation.

Physics & Robotics Instructor | 2007 – 2013

- Taught physics, robotics, and scientific concepts to students.
- Supervised technical and educational projects, including robotics and problem-solving activities.
- Developed strong communication, teaching, analytical thinking, and project guidance skills.

Independent Learning & Technical Project Development | 2021 – Present

- Continued developing skills in Python, AI, data analysis, computer vision, and software engineering.
- Built practical portfolio projects and strengthened programming foundations through 42 Heilbronn and self-directed learning.

EDUCATION

42 Heilbronn – Software Engineering | 2025 – Present – Project-based software engineering program focused on C programming, Linux, Git, algorithms, problem solving, peer learning, and practical coding projects.

M.Sc. Artificial Intelligence – Amirkabir University of Technology

B.Sc. Physics – University of Tehran

LANGUAGES

Persian – Native | English – Fluent | German – Upper-intermediate

STRENGTHS

Analytical Thinking | Problem Solving | Fast Learning | Structured Working Style | Teamwork | Initiative | Reliability

INTERESTS

Astronomy | Artificial Intelligence | Robotics | Science Communication | Hiking | Yoga | driving